

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3-AT2 - DATA INTERPRETATION

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3 – Data Integration	Assessment:	Assessment Tool 2– Data Integration
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Apply statistical analysis techniques to summarize data, detect outliers, measure asymmetry, and compare variability for accurate interpretation of real-world datasets. (BL4)		
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Section / Batch:	2024	Date:	18.08.2026

Code of Conduct

I **Jeya Harshini R (192424051)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jeya Harshini R

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1: Data Summarization

Given Dataset (Employee Salaries in ₹'000)

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

(a) Compute Mean and Median

Mean

Formula:

$$Mean = \frac{\sum X}{n}$$

Where:

$$\sum X = 32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120$$

$$\sum X = 581$$

Number of observations:

$$n = 11$$

Therefore,

$$Mean = \frac{581}{11}$$

$$Mean = 52.82$$

Mean

Salary = ₹52.82 thousand

Median

Since the data is already arranged in ascending order and there are 11 observations:

$$\begin{aligned} \text{Median} &= \frac{n + 1}{2} \\ \text{Median} &= \frac{11 + 1}{2} \\ \text{Median} &= 6^{\text{th}} \text{ Observation} \end{aligned}$$

6th observation = 42

Median Salary = ₹42 thousand

(b) Why is Mean a Poor Measure of Central Tendency?

The mean salary is ₹52.82 thousand, whereas most employee salaries lie between ₹32 thousand and ₹50 thousand.

The two unusually large salaries (₹95 thousand and ₹120 thousand) pull the mean upward, making it unrepresentative of the majority of employees.

The median remains unaffected by these extreme values and better represents the central tendency of the dataset.

Therefore, the mean is a poor measure because it is highly sensitive to outliers.

(c) Standard Deviation or IQR?

The Interquartile Range (IQR) is preferred.

Reason:

- Standard deviation is affected by extreme values.
- IQR focuses only on the middle 50% of the data.
- IQR is resistant to outliers.
- Since the dataset contains unusually high salaries, IQR provides a more reliable measure of dispersion.

Conclusion

For this dataset, **IQR is more trustworthy than standard deviation** because it is robust to outliers.

Question 2: Outlier Treatment Using $1.5 \times \text{IQR}$ Rule

Dataset

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

Step 1: Find Q1 and Q3

Median = 42

Lower Half:

32, 35, 36, 38, 40

Q1 = 36

Upper Half:

45, 48, 50, 95, 120

Q3 = 50

Step 2: Calculate IQR

$$IQR = Q3 - Q1$$

$$IQR = 50 - 36$$

$$IQR = 14$$

Step 3: Calculate Fences

Lower Fence

$$Q1 - 1.5(IQR)$$

$$36 - 1.5(14)$$

$$36 - 21$$

$$15$$

Upper Fence

$$Q3 + 1.5(IQR)$$

$$50 + 1.5(14)$$

$$50 + 21$$

$$71$$

Step 4: Identify Outliers

Any value:

- Less than 15
- Greater than 71

Outliers:

95, 120

Outliers Identified: ₹95 thousand and ₹120 thousand

Effect of Winsorization

If outliers are capped at the upper fence (71) instead of being removed:

Original outliers:

95 → 71

120 → 71

This reduces the impact of extreme values.

Qualitative Impact

- Mean decreases significantly.
- Dataset becomes less skewed.
- Central tendency becomes more representative.
- Variability reduces.

Thus, Winsorization improves robustness while retaining all observations.

Question 3: Measuring Asymmetry

Two datasets have:

- Mean = 50
 - Standard Deviation = 10
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Dataset A

Pearson Skewness = +1.2

Shape

- Positively skewed distribution
- Long tail on the right side
- Majority of observations concentrated on the left
- Few extremely large values

Graphically:

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Interpretation

Most values are lower, with a few unusually large observations.

Dataset B

Pearson Skewness = -0.8

Shape

- Negatively skewed distribution
- Long tail on the left side
- Majority of observations concentrated on the right
- Few extremely small values

Graphically:

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Interpretation

Most values are high, with a few unusually low observations.

Which Dataset Represents Salary Data?

Dataset A.

Reason

Salary distributions typically exhibit:

- Many employees earning moderate salaries
- Few employees earning very high salaries
- Long right tail

Hence, Dataset A with skewness +1.2 is more representative of salary data.

Question 4: Coefficient of Skewness Calculation

Given

Mean = 62

Median = 58

Mode = 52

Standard Deviation = 12

(a) Karl Pearson's Coefficient (Mode Method)

Formula:

$$Sk = \frac{Mean - Mode}{SD}$$

Substituting values:

$$Sk = \frac{62 - 52}{12}$$

$$Sk = \frac{10}{12}$$

$$Sk = 0.833$$

Karl Pearson Skewness (Mode Method)

$$Sk = 0.833$$

(b) Karl Pearson's Coefficient (Median Method)

Formula:

$$Sk = \frac{3(Mean - Median)}{SD}$$

Substituting values:

$$Sk = \frac{3(62 - 58)}{12}$$

$$Sk = \frac{3(4)}{12}$$

$$Sk = \frac{12}{12}$$

$$Sk = 1.0$$

Karl Pearson Skewness (Median Method)

$$Sk = 1.0$$

Comparison

Method	Skewness
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Mode Method	0.833
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Median Method	1.000
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Comment

Both coefficients are positive, indicating a positively skewed distribution.

The values differ slightly because:

- The mode may be difficult to estimate accurately.
- Mean, median, and mode may not perfectly satisfy the empirical relationship.

Despite the numerical difference, both methods indicate moderate positive skewness.

Question 5: Variance vs Standard Deviation Interpretation

Given

Line A

Mean = 200 units/day

Variance = 64

$$SD = \sqrt{64}$$

$$SD = 8$$

Line B

Mean = 150 units/day

Variance = 36

$$SD = \sqrt{36}$$

$$SD = 6$$

Coefficient of Variation

Formula:

$$CV = \frac{SD}{Mean} \times 100$$

Line A

$$CV_A = \frac{8}{200} \times 100$$
$$CV_A = 4\%$$

Line B

$$CV_B = \frac{6}{150} \times 100$$
$$CV_B = 4\%$$

Comparison

Production Line Mean SD CV

Line A	200	8	4%
Line B	150	6	4%

Interpretation

Although Line A has a larger variance (64) than Line B (36), variance alone cannot be used to compare variability because the means are different.

The Coefficient of Variation standardizes variability relative to the mean.

Since both lines have the same CV of 4%, they exhibit identical relative variability.

Conclusion

This assignment explored important statistical concepts including measures of central tendency, dispersion, outlier detection, skewness analysis, and relative variability. The results demonstrate that robust measures such as the median and IQR are preferable in the presence of outliers, skewness helps identify distribution shape, and the coefficient of variation enables meaningful comparison of variability across datasets with different means. These techniques are essential for accurate data interpretation and decision-making in real-world applications.

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